ROHIT THAKUR

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AI/ML engineer with 8+ years building and running production AI systems across computer vision, generative AI, and manufacturing/IoT analytics. Builds agentic LLM systems (RAG, LangGraph, tool use) with guardrails and measured evaluation, and real-time on-device vision (detection, facial landmarks, pose, segmentation) tuned for edge hardware with ONNX and quantization. Recent work delivered 20% higher manufacturing efficiency, a 33% reduction in prediction error, and 95.5% fault-detection accuracy. Comfortable owning the whole pipeline, from data and training through deployment and monitoring, in Agile (SAFe) delivery.

TECHNICAL SKILLS

Programming	Python, C++, SQL, R, Java
Generative AI & LLM	Agentic RAG, LangGraph, LangChain, hybrid retrieval (dense + BM25), vector databases (Chroma, FAISS), prompt engineering, Model Context Protocol (MCP), LLM evaluation (LLM-as-Judge, DeepEval, NLI-based faithfulness and hallucination detection), uncertainty quantification and calibration, guardrails and prompt-injection defense, AWS Bedrock (Claude), OpenAI GPT-4o, Hugging Face Transformers
Computer Vision	OpenCV, YOLOv5/v8, U-Net, object detection, facial landmarks, head-pose estimation, image segmentation, DMS/OMS/ADAS, MediaPipe
Deep Learning & Edge AI	PyTorch, TensorFlow, Keras, scikit-learn, ONNX, ONNX Runtime, INT8 quantization, pruning, TensorRT, model lightweighting, multi-task learning
Data & Cloud	Snowflake, Snowpark, Pandas, NumPy, SciPy, Spark, Kafka, Hadoop, Azure ML, Databricks
MLOps & DevOps	Docker, Kubernetes, MLflow, Weights and Biases, Git, CI/CD, FastAPI, Langfuse, Streamlit
Visualization	Plotly, Matplotlib, Altair, React (data dashboards)

PROFESSIONAL EXPERIENCE

AI/ML Engineer — eMoldino, Seoul, South Korea
Apr 2025 – Present

- Built a Production Insight LLM agent (AWS Bedrock Claude 3.7 Sonnet, LangGraph, Snowflake) with intent-driven routing and real-time streaming to a React frontend.
- Designed a Manufacturing MCP Server (v3.1) exposing four analytics tools (ROI, RunRate, Capacity, CT Efficiency) over FastAPI with async job queues and multi-user sessions, plus a Tooling Report agent (LangChain) that writes executive summaries and graded recommendations.
- Built an end-to-end LLM evaluation suite (emoldino-eval): 210+ tests across four manufacturing modules using LLM-as-Judge and DeepEval metrics (faithfulness, hallucination, answer relevancy) over six scored criteria, with results tracked in Snowflake.

- Developed an AI-powered cycle-time deviation tool with adaptive windows and Mann-Whitney U / Chi-square testing, and a capacity-risk dashboard using Physics-of-Loss methodology, rolling risk scoring, OLS forecasting, and Bedrock-generated insights with Langfuse observability.
- Built sensor-data pipelines and anomaly detection: downtime detection from time-gap analysis with 11+ engineered features, and acceleration-sensor RMS/Crest Factor with Kolmogorov-Smirnov testing and cosine-similarity analysis, on automated Snowflake ingestion.
- Deployed a scrap-rate prediction pipeline that cut test MAE by 33% (16.3% to 10.9%).

AI Engineering Manager — OpenSysNet Co., Ltd., Pangyo, South Korea May 2022 – Apr 2025

- Ran AI delivery under the Scaled Agile Framework (SAFe), improving sprint velocity and release consistency.
- Developed a real-time railway pantograph monitoring system with YOLOv8 and edge processing, reaching 95.5% fault-detection accuracy; received an Innovation Award for it.
- Designed a sorting-robot vision pipeline at 99.5% mAP@50 that raised manufacturing efficiency by 20%.
- Built LSTM and gradient-boosting models for fan-speed control and energy-anomaly detection, saving 12.3% in energy.
- Developed computer-vision applications for golf-swing analysis, face/fire/hand detection, and OCR from drone video feeds.

AI Research & Development Engineer — Pittasoft Co., Ltd., Seoul, South Korea Sep 2017 – Apr 2022

- Led ADAS development for dashcam products: forward vehicle detection, traffic-light recognition, and Forward Vehicle Start Alarm, using OpenCV and custom algorithms.
- Built a Driver Monitoring System (DMS), training TensorFlow YOLO models to detect distraction and drowsiness.
- Applied a VGG-based U-Net to medical image segmentation with optimized real-time inference.
- Built end-to-end video-processing pipelines (OpenCV, FFmpeg) for real-time streaming, image enhancement, and feature matching.

SELECTED PROJECTS

CabinSense — Multi-Task In-Cabin Driver and Occupant Monitoring (Independent)

- Real-time DMS/OMS inferring drowsiness, distraction, gaze zone, yawning, phone use, smoking, seatbelt, glasses, cap, and occupant count by fusing ONNX face detection, 98-point facial landmarks, head-pose geometry (EAR/MAR/PERCLOS), and a trained MobileNetV3 multi-task classifier.
- Engineered for edge/SoC deployment: ONNX export with INT8 quantization, a model manifest, temporal-stability fusion, and a documented C/C++ portability path. PyTorch, ONNX Runtime, MLflow, seeded and unit-tested.

Agentic RAG with Guardrails (Independent)

- Fully local agentic RAG assistant (LangGraph) over enterprise documents with hybrid retrieval (dense + BM25), NLI-based hallucination and groundedness guardrails, prompt-injection defense, and document-level access control.
- Implemented a judge-free retrieval and answer-quality harness (hit@k, MRR, nDCG, NLI faithfulness) and a two-model accuracy/latency/cost comparison; served via FastAPI with API-key auth and monitoring.

Uncertainty Quantification for LLMs (Independent)

- Estimated token- and sentence-level uncertainty for a local open LLM (Qwen2.5-1.5B): predictive entropy, perplexity, max-softmax confidence, and semantic entropy via NLI meaning-clustering, benchmarked on factual-QA and math tasks.
- Measured usefulness with error-detection AUROC, calibration (ECE, reliability diagrams), and risk-coverage curves; added uncertainty-aware selective answering and self-consistency decoding. PyTorch and Hugging Face, fully local and seeded.

EDUCATION

M.S., Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology (GIST), South Korea, 2017. Specialization: artificial intelligence and machine learning.

B.Tech., Electronics and Communication Engineering, Green Hills Engineering College, India, 2014.

PUBLICATIONS

- "Energy Consumption Prediction Using LSTM Models," International Conference on Machine Learning and Applications (ICMLA).
- "Concurrent Food Localization and Recognition," Korean Institute of Communications and Information Sciences (KICS), 2017.
- "Augmentation of RGB Dataset for Improved Performance on Infrared Classifier," Electronics and Information Communications Conference, 2016.

ACHIEVEMENTS & LANGUAGES

- Innovation Award (OpenSysNet) for the pantograph monitoring system, 95.5% fault-detection accuracy.
- Employee of the Year (Pittasoft) for work on ADAS and driver monitoring systems.
- IEEE member, active in AI/ML and computer-vision communities.
- English (fluent), Korean (professional working proficiency), Hindi (native).